

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14 COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: *Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:100

Code of Conduct:

I __M Kavya 192472370__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: kavya

Q1 — scanme.nmap.org

Objective:

To perform initial enumeration of the publicly accessible server using Nmap and identify open ports, services, service versions, operating system information, and default script results.

Commands Used:

`nmap -sn scanme.nmap.org`

`sudo nmap -sS scanme.nmap.org`

`sudo nmap -p- scanme.nmap.org`

`sudo nmap -sV scanme.nmap.org`

`sudo nmap -O scanme.nmap.org`

`sudo nmap -sC scanme.nmap.org`

`sudo nmap -sS -sV -O -sC scanme.nmap.org`

Record in the document:

- Host status
- Open TCP ports
- Service names
- Service versions
- OS information
- NSE/default script results

Analysis:

Explain what each discovered service is used for and whether it needs to be publicly accessible. Mention that exposed services increase the attack surface and may provide information useful for vulnerability discovery, service exploitation, or credential attacks.

Recommendations:

- Disable unnecessary services.
- Keep exposed services updated.
- Restrict administrative services.
- Use firewall rules.
- Monitor exposed ports and services.
- Apply strong authentication and rate limiting where applicable.

```
(kavya@kavya)-[~]
$ sudo nmap -sS -sV -O -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 19:08 +0530
Stats: 0:01:05 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 43.01% done; ETC: 19:11 (0:01:25 remaining)
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).
Stats: 0:01:59 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 43.83% done; ETC: 19:13 (0:02:33 remaining)
Stats: 0:04:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.90% done; ETC: 19:18 (0:05:06 remaining)
Stats: 0:06:23 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 46.01% done; ETC: 19:22 (0:07:29 remaining)
Stats: 0:09:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 47.44% done; ETC: 19:28 (0:10:16 remaining)
Stats: 0:17:25 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 51.47% done; ETC: 19:42 (0:16:25 remaining)
Stats: 0:32:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 58.69% done; ETC: 20:03 (0:22:31 remaining)
Stats: 0:33:51 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 59.60% done; ETC: 20:05 (0:22:57 remaining)
Stats: 0:37:50 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 61.58% done; ETC: 20:10 (0:23:36 remaining)
Stats: 0:39:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 62.31% done; ETC: 20:12 (0:23:46 remaining)
Stats: 0:40:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 62.76% done; ETC: 20:13 (0:23:52 remaining)
Stats: 0:42:34 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 64.01% done; ETC: 20:15 (0:23:55 remaining)
```

Q2 — Metasploitable

Objective:

To comprehensively enumerate the intentionally vulnerable Linux server and identify services that may act as potential entry points.

Commands Used:

```
nmap -sn 192.168.56.101
sudo nmap 192.168.56.101
sudo nmap -p- 192.168.56.101
sudo nmap -sV 192.168.56.101
sudo nmap -O 192.168.56.101
sudo nmap -sC 192.168.56.101
sudo nmap -sS -sV -O -sC -p- 192.168.56.101
```

Record in the document:

- Number of open ports
- Port numbers
- Protocols
- Service names
- Versions
- OS detection result
- Interesting NSE findings

Risk Classification:

Risk	What to mention
High	Outdated/vulnerable services, insecure remote-access services, services allowing unauthorized access
Medium	Services that expose useful information or provide additional attack surface
Low	Services with limited exposure or low-impact information disclosure

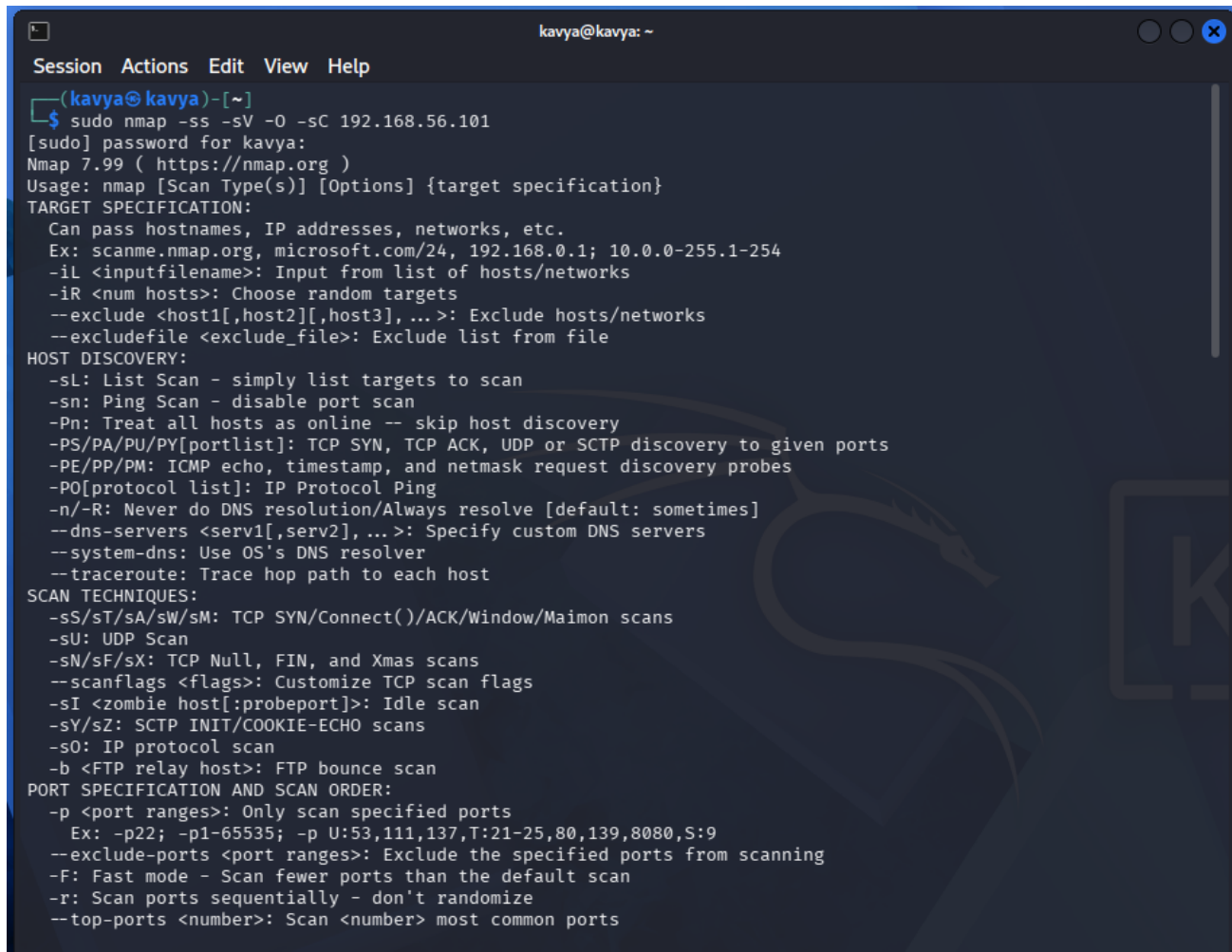
Analysis:

Explain how enumeration helps a penetration tester identify possible attack paths. For example, an outdated service version can be compared against known vulnerabilities, while exposed authentication services may be assessed for weak credentials.

Recommendations:

- Remove unnecessary services.
- Patch outdated software.

- Disable insecure protocols.
- Apply firewall restrictions.
- Use strong authentication.
- Segment vulnerable laboratory systems.



```

kavya@kavya: ~
Session Actions Edit View Help
(kavya@kavya)-[~]
$ sudo nmap -ss -sV -O -sC 192.168.56.101
[sudo] password for kavya:
Nmap 7.99 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports

```

Q3 — Windows Workstation

Objective:

To enumerate the Windows workstation and identify file-sharing, remote-management, and other network services that could increase the attack surface.

Commands Used:

```
nmap -sn 192.168.56.102
```

```
sudo nmap 192.168.56.102
```

```
sudo nmap -p- 192.168.56.102
```

```
sudo nmap -sV 192.168.56.102
```

```
sudo nmap -O 192.168.56.102
```

```
sudo nmap -sC 192.168.56.102
```

```
sudo nmap --script smb-enum-shares,smb-enum-users -p 139,445 192.168.56.102
```

```
sudo nmap -sV -p 135,139,445,3389,5985,5986 192.168.56.102
```

Important ports to check:

Port	Service	Purpose
135	RPC	Windows RPC communication
139	NetBIOS/SMB	File/printer sharing
445	SMB	Windows file and network sharing
3389	RDP	Remote desktop
5985	WinRM	Remote Windows management
5986	WinRM HTTPS	Secure remote management

Analysis:

Discuss whether these services are actually required. Services that are unnecessarily exposed violate the principle of least privilege and increase the system's attack surface.

Possible attack implications:

- SMB can reveal shared resources.
- RDP can become a target for unauthorized remote access.
- RPC/NetBIOS can provide information useful for reconnaissance.
- Remote-management services can potentially facilitate lateral movement if credentials are compromised.

Recommendations:

- Disable unused services.
- Restrict SMB access using firewall rules.
- Disable unnecessary RDP.
- Use strong authentication.
- Apply network segmentation.
- Keep Windows and services patched.

```
kavya@kavya: ~  
Session Actions Edit View Help  
(kavya@kavya)-[~]  
$ sudo nmap -sV -p 135,139,445,3389,5985,5986 192.168.56.102  
[sudo] password for kavya:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 19:23 +0530  
Nmap scan report for 192.168.56.102  
Host is up (0.00069s latency).  
  
PORT      STATE      SERVICE      VERSION  
135/tcp    filtered  msrpc  
139/tcp    filtered  netbios-ssn  
445/tcp    filtered  microsoft-ds  
3389/tcp   filtered  ms-wbt-server  
5985/tcp   filtered  wsman  
5986/tcp   filtered  wsmans  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.07 seconds
```

Q4 — OWASP Juice Shop

Objective:

To identify the network exposure of the Juice Shop server and determine which services support the web application.

Commands Used:

```
nmap -sn 192.168.56.103
```

```
sudo nmap 192.168.56.103
```

```
sudo nmap -p- 192.168.56.103
```

```
sudo nmap -sV 192.168.56.103
```

```
sudo nmap -O 192.168.56.103
```

```
sudo nmap -sC 192.168.56.103
```

```
sudo nmap -sV -p 80,443,3000,8080,8000,8443 192.168.56.103
```

```
sudo nmap --script http-title,http-headers,http-methods -p 80,443,3000,8080,8000,8443 192.168.56.103
```

Record:

- Web ports
- HTTP/HTTPS services
- Web-server/application information
- Service versions
- OS information
- HTTP titles/headers/methods

Analysis:

Explain how exposed web services provide the entry point to the application. Information such as server versions, HTTP methods, headers, and exposed administrative interfaces can help testers identify areas requiring further security testing.

Recommendations:

- Expose only required web ports.
- Use HTTPS.
- Remove unnecessary administrative interfaces.
- Keep the web server and dependencies updated.
- Configure secure HTTP headers.
- Use firewall rules and network segmentation.
- Apply authentication and authorization controls.

```
(kavya@kavya)~$ sudo nmap -sV -p 3000 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 19:23 +0530
Nmap scan report for 192.168.56.103
Host is up (0.00030s latency).

PORT      STATE      SERVICE VERSION
3000/tcp   filtered   ppp

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.98 seconds
```

Q5 — Gateway 192.168.56.1

Objective:

To identify exposed management interfaces and network services on the gateway and evaluate their security implications.

Commands Used:

```
nmap -sn 192.168.56.1
sudo nmap 192.168.56.1
sudo nmap -p- 192.168.56.1
sudo nmap -sV 192.168.56.1
sudo nmap -O 192.168.56.1
sudo nmap -sC 192.168.56.1
sudo nmap -sV -p 22,23,80,443,161,162 192.168.56.1
sudo nmap -sU -p 161 --script snmp-info 192.168.56.1
sudo nmap -sS -sV -O -sC -p- 192.168.56.1
```

Important services:

Port	Service	Security concern
22	SSH	Remote administration; should be restricted
23	Telnet	Unencrypted and insecure
80	HTTP	Unencrypted management interface

443	HTTPS	Encrypted management interface
161	SNMP	May expose device/network information
162	SNMP Trap	Monitoring/management traffic

Analysis:

Explain that management interfaces should generally not be exposed broadly. Telnet is particularly risky because credentials and traffic can be transmitted without encryption. SNMP may disclose useful network information if poorly configured.

Recommendations:

- Disable Telnet.
- Prefer SSH over insecure remote-management protocols.
- Restrict management access to trusted administrator networks.
- Use HTTPS for web administration.
- Use strong SNMP configuration and credentials.
- Apply ACLs/firewall rules.
- Keep gateway firmware updated.
- Enable logging and monitoring.

```
(kavya@kavya)-[~]
$ sudo nmap -sU --top-ports 100 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 19:24 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00045s latency).
Not shown: 94 filtered udp ports (port-unreach)
PORT      STATE      SERVICE
123/udp    open|filtered ntp
137/udp    open|filtered netbios-ns
138/udp    open|filtered netbios-dgm
1900/udp   open|filtered upnp
4500/udp   open|filtered nat-t-ike
5353/udp   open|filtered zeroconf

Nmap done: 1 IP address (1 host up) scanned in 3.25 seconds
```

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
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Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	Develops accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	